3D Data Science with Python

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3D used to mean C++. Not anymore

C++ with PCL

```
using Cloud =
    pcl::PointCloud<pcl::PointXYZ>;
Cloud::Ptr cloud(new Cloud);
Cloud down;
if (pcl::io::loadPCDFile(
    "crop.pcd", *cloud)) return 1;
pcl::VoxelGrid<pcl::PointXYZ> vg;
vg.setInputCloud(cloud);
vg.setLeafSize(0.5f, 0.5f, 0.5f);
vg.filter(down);
std::cout << cloud->size()
  << " -> " << down.size() << "\n";


# output
722961 -> 230519
```

C++ output: recorded Linux / PCL run; not rerun on this machine.

Python with Open3D

```
pcd = o3d.io.read_point_cloud(path)
down = pcd.voxel_down_sample(0.5)
print(len(pcd.points), "->",
      len(down.points))


# output
722961 -> 232180
```

Why the counts differ: both average the points in each 0.5 m voxel, but the grids sit in different places. PCL anchors its voxels at multiples of the leaf size; Open3D at the minimum corner minus half a voxel.

Python is the interface. C++ does the work



Every call you make here runs compiled code

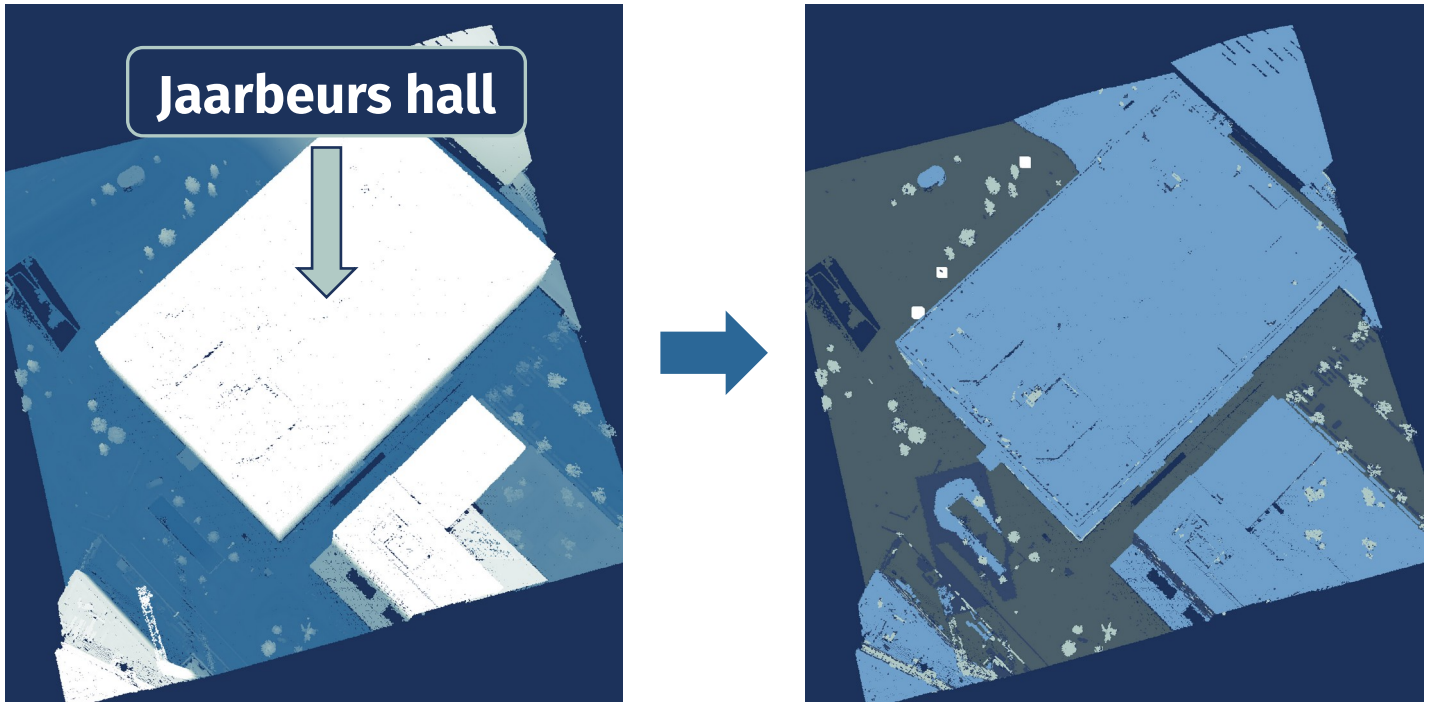
- NumPy: a compiled C core, with some C++
- Open3D: C++, wrapped for Python
- PyVista: VTK, which is C++
- laspy: the lazrs decoder is Rust

Python is used as a proxy to write higher-level code: the heavy loops run below it, in compiled code.

The Python 3D ecosystem

Load and store	laspy	PDAL	pyproj	
Arrays and speed	NumPy	SciPy	Numba	dask
Process	OPEN3D	trimesh		
Visualise	PyVista	VTK		
Learn	scikit-learn	PyTorch3D		

Today's case: the Jaarbeurs hall, from a plane



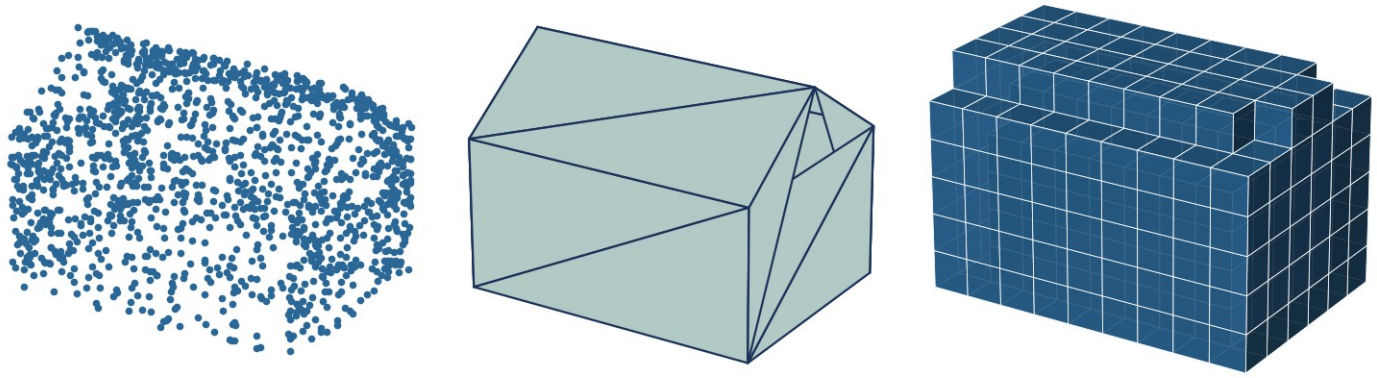
The data set

Area	200 m × 200 m
Points	722 961
Density	about 18 per m²
Source	AHN5, ahn.nl
Flown	April 2023, night
File	LAZ, 4.3 MB
Coordinates	RD New + NAP
Licence	CC BY 4.0

Goal: ground, buildings, vegetation, street furniture; scored against AHN.

Data: AHN5, Het Waterschapshuis, CC BY 4.0.

One object, three representations



Point cloud

1500 × 3 floats
no connectivity

Mesh

10 vertices, 16 triangles
a surface with neighbours

Voxel grid

182 occupied 0.5 m cells
volume at one resolution

Points measure, meshes connect, voxels index space. Convert only when the task needs it.

- Point cloud: an unordered N×3 array plus attributes; no neighbours stored.
- Mesh: vertices plus faces, so surfaces, normals and areas are defined.
- Voxel grid: occupied cells at one resolution; detail below a cell is gone.

A 3D project moves through six stages



Acquisition
Scan the scene
laspy

1

2

Preparation
Clean and register
NumPy, Open3D

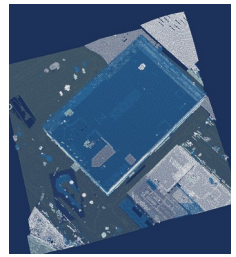
Application
Ship it

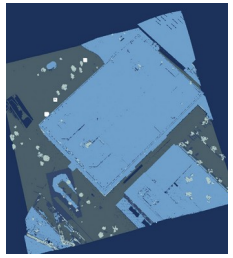
6

Loop back when the data surprises

3

Exploration
Model and label
Open3D, SciPy





Visualisation
See it, trust it
PyVista

5

4

Analysis
Measure, monitor
NumPy, SciPy

Data: AHN5, Het Waterschapshuis, CC BY 4.0.

Four rules of thumb for 3D data

1 Think in arrays, not points

A point cloud is an N×3 table plus attribute columns.

2 Look before you compute

Render it; check units, coordinates and classes.

3 Make it small first

Crop, tile and downsample; scale up last.

4 Let compiled code loop

One call per array or tile, never per point.

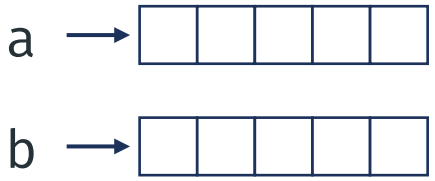
No single library does the whole job

Load	Voxel downsampling	Ground removal	Clustering	Labelling	Visualisation
laspy LAS/LAZ in and out	**Open3D** downsampling, planes, DBSCAN				**PyVista** off-screen renders
			SciPy KDTree batch neighbour queries		
NumPy the arrays every step passes on					

Use views, not copies

An extra xyz copy costs 24 B per point → 0.5 GiB per 1 km file.

The problem: copies `copy`



new memory → costs time & RAM

The solution: views `view`

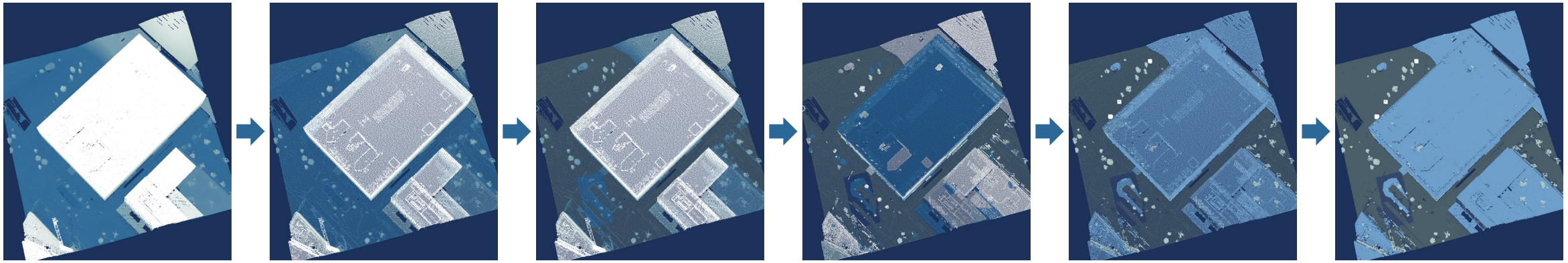


same memory → free, writes show through

`las.xyz`	copy	new array, every access
`Vector3dVector(xyz)`	copy	float32: 39× slower
`np.asarray(pcd.points)`	view	writes reach the point cloud
`pv.PolyData(pts)`	view	same memory as pts

Conditions: Ryzen 7 5700X, Python 3.13.15, NumPy 2.5.3; 722 961 points, median of 7 runs; AHN5 crop (CC BY 4.0). Full: results/python_facts.json.

Six steps turn a point cloud into labels



Load	**Voxel downsampling**	**Ground removal**	**Clustering**	**Labelling**	**Visualisation**
laspy	Open3D	Open3D	Open3D, SciPy	NumPy, SciPy	SciPy, PyVista

Data: AHN5, Het Waterschapshuis, CC BY 4.0.

1 Load: laspy hands you NumPy

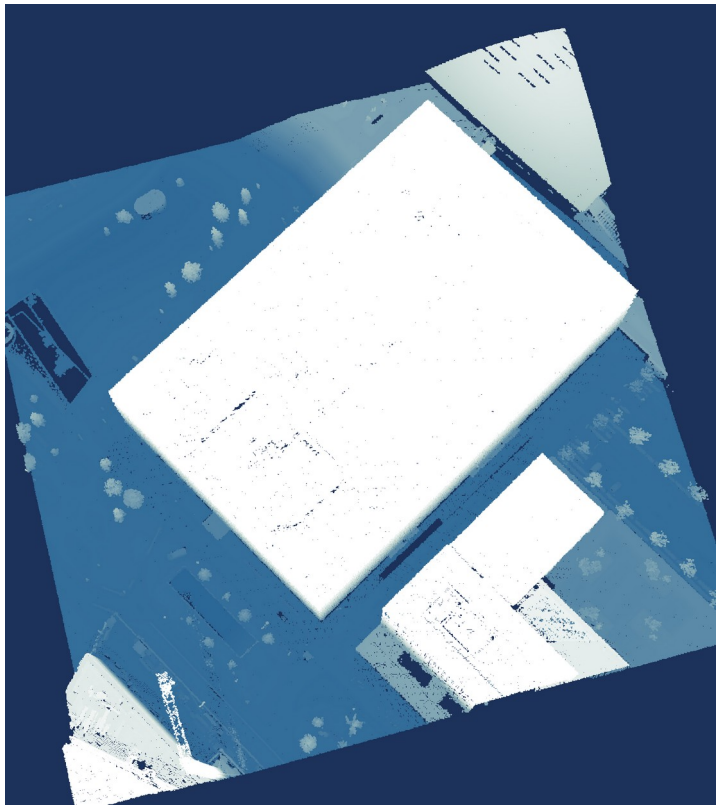
```python
import laspy
import open3d as o3d
las = laspy.read("crop.laz")
xyz = las.xyz - las.header.mins
v = o3d.utility.Vector3dVector(xyz)
pcd = o3d.geometry.PointCloud(v)
```



float32 step

1.6 cm →
after origin shift:

15 µm

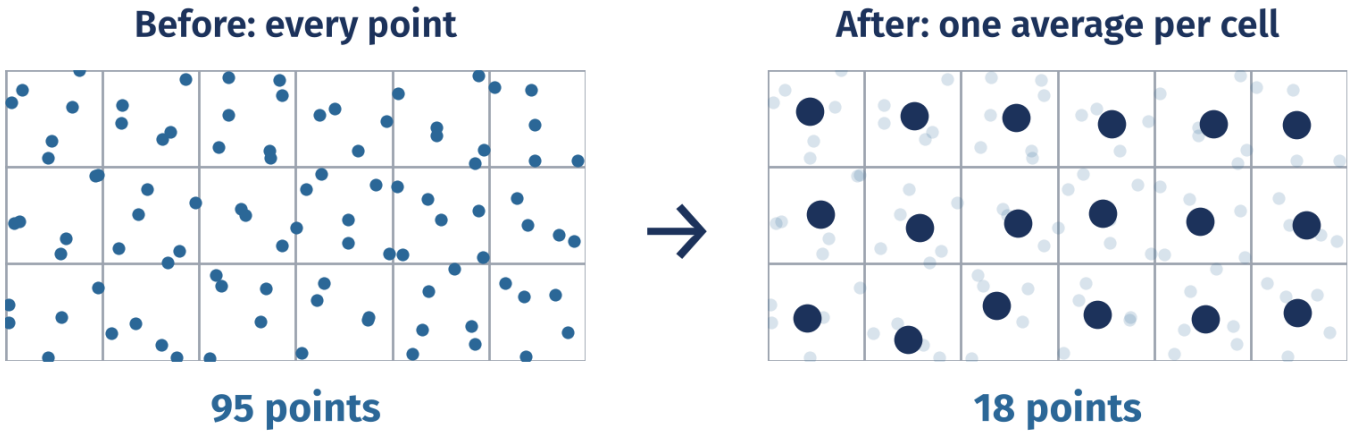


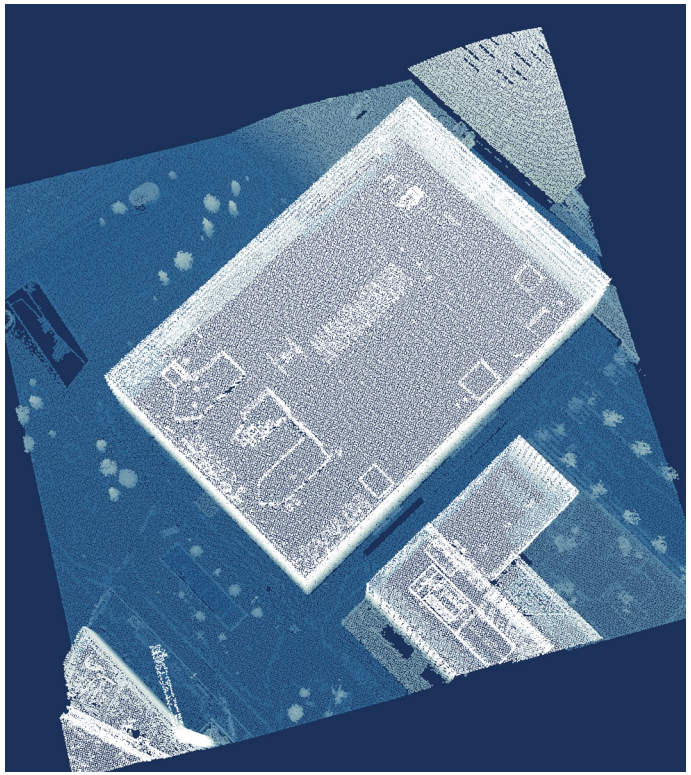
Subtract a local origin before anything else.

Data: AHN5, Het Waterschapshuis, CC BY 4.0.

2 Voxel downsampling: one average per voxel

```
down = pcd.voxel_down_sample(0.5)
P = np.asarray(down.points)
print(len(xyz), "->", len(P))
```





722 961 points in → 232 180 out at 0.5 m voxels.

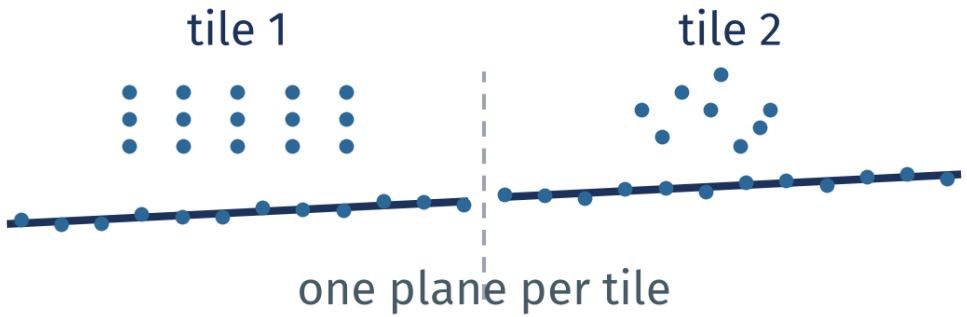
Data: AHN5, Het Waterschapshuis, CC BY 4.0.

3 Ground removal: one plane per tile

```
fits = []
for i in tiles(P, 25):
    z0 = np.percentile(P[i, 2], 2)
    low = i[P[i, 2] < z0 + 1.5]
    t = down.select_by_index(low)
    plane, inl = t.segment_plane(
        0.4, 3, 300,
        probability=1.0)
    fits += [(i, low, inl, plane)]
# Plane too steep or > 2.5 m above
# the median? Use nearest good one.
ground, _ = check_planes(P, fits)
```



Vs AHN's ground class:
98.4 % precision,
94.4 % recall.

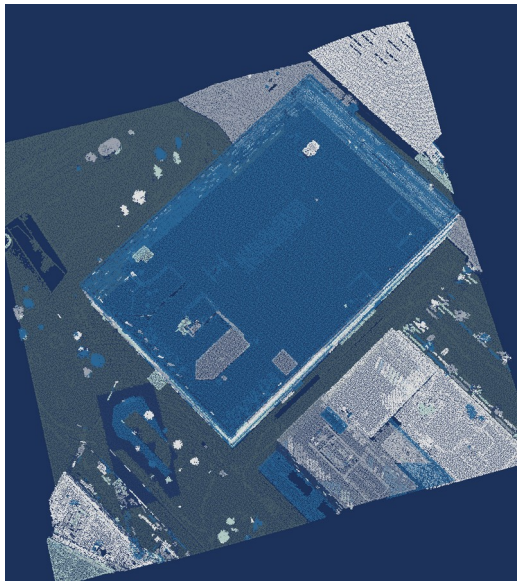


**One plane per tile,
checked against its neighbours.**

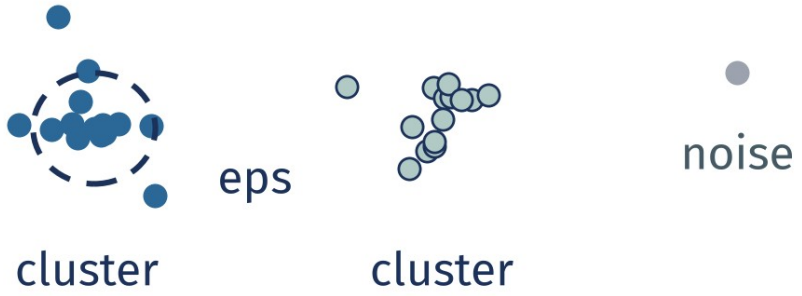
Data: AHN5, Het Waterschapshuis, CC BY 4.0.

4 Clustering: DBSCAN groups what touches

```
hag = height_above(P, ground)
up = ~ground & (hag > 0.3)
rest = down.select_by_index(
    np.flatnonzero(up))
lab = rest.cluster_dbscan(
    eps=0.75, min_points=6)
ids = np.asarray(lab)
print(ids.max() + 1, "clusters")
```



1 026 clusters.
Clusters are not labels yet.

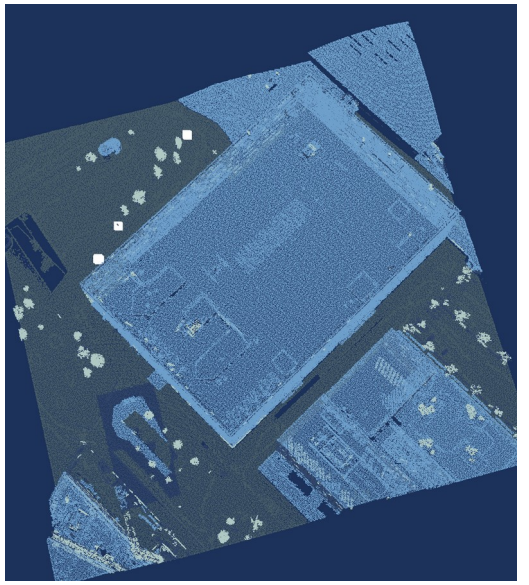


Neighbours within eps make a cluster.

Data: AHN5, Het Waterschapshuis, CC BY 4.0.

5 Labelling: rules name the clusters

```
def name(c):
    if c.multi > 0.4 and c.top > 1:
        return VEGETATION
    if c.foot > 25 and c.top > 2.5:
        return BUILDINGS
    if c.to_building < 2:  # facade
        return BUILDINGS
    if c.slender and c.free:
        return STREET_FURNITURE
    return UNLABELLED
```



Leaf-off scan: bare trees echo several times.

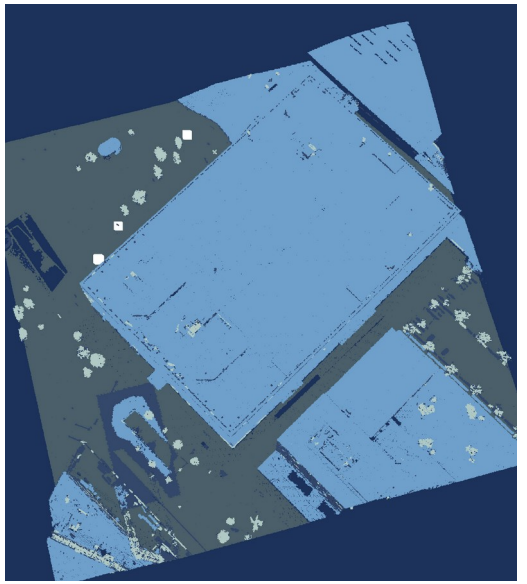


The first rule that fires names the cluster.

Data: AHN5, Het Waterschapshuis, CC BY 4.0.

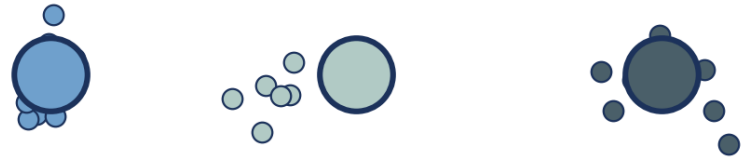
6 Visualisation: PyVista paints every point

```
P = np.asarray(down.points)
_, j = KDTree(P).query(
    xyz, workers=-1)
pl = pv.Plotter(off_screen=True)
pl.add_points(xyz, point_size=3,
    scalars=RGB[cls[j]], rgb=True)
pl.camera_position = CAM
pl.screenshot("f6_final.png")
```



Fixed camera, off-screen, one PNG per step.

Paint the labels back onto every point.

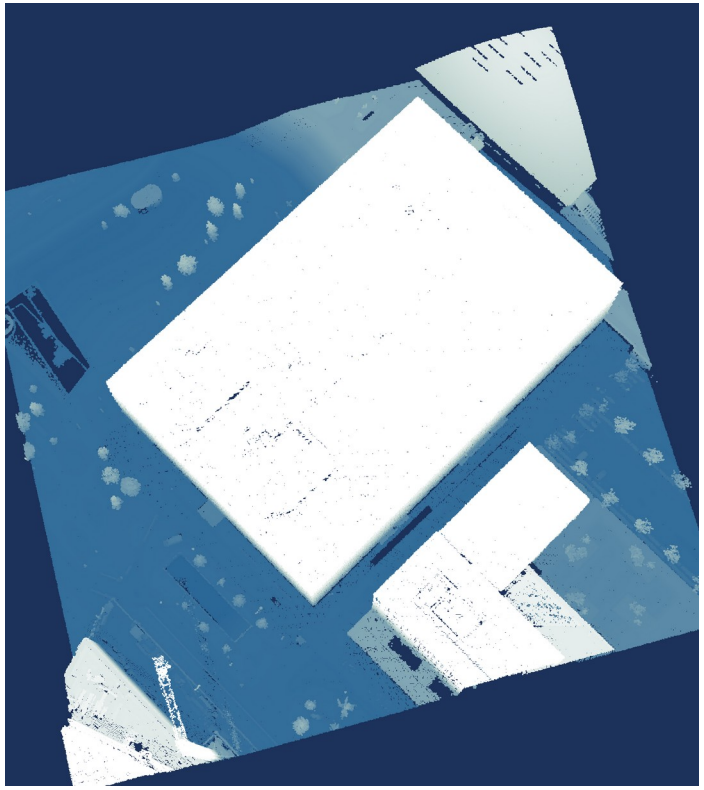


each point takes its nearest voxel's colour

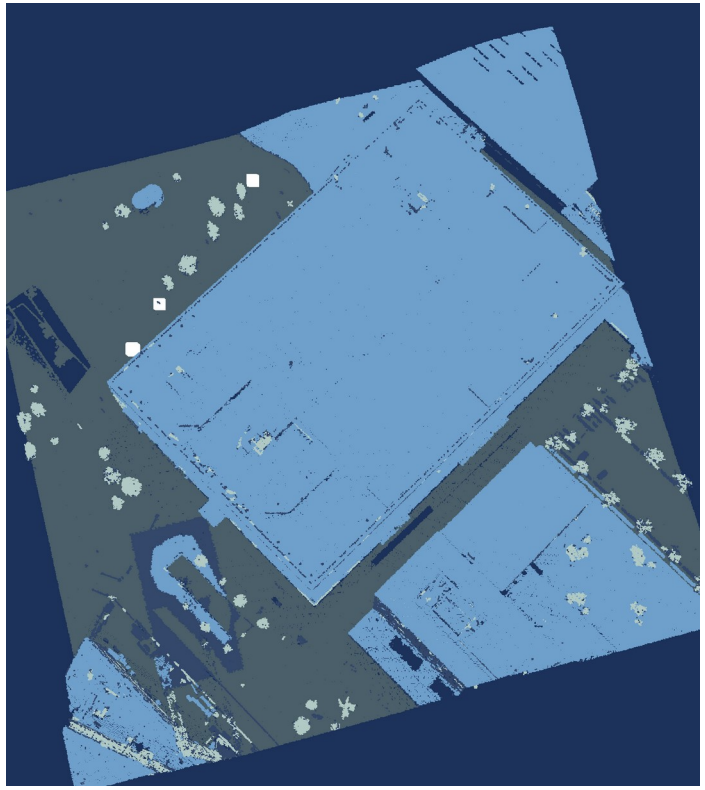
Data: AHN5, Het Waterschapshuis, CC BY 4.0.

What the six steps did

Original scan



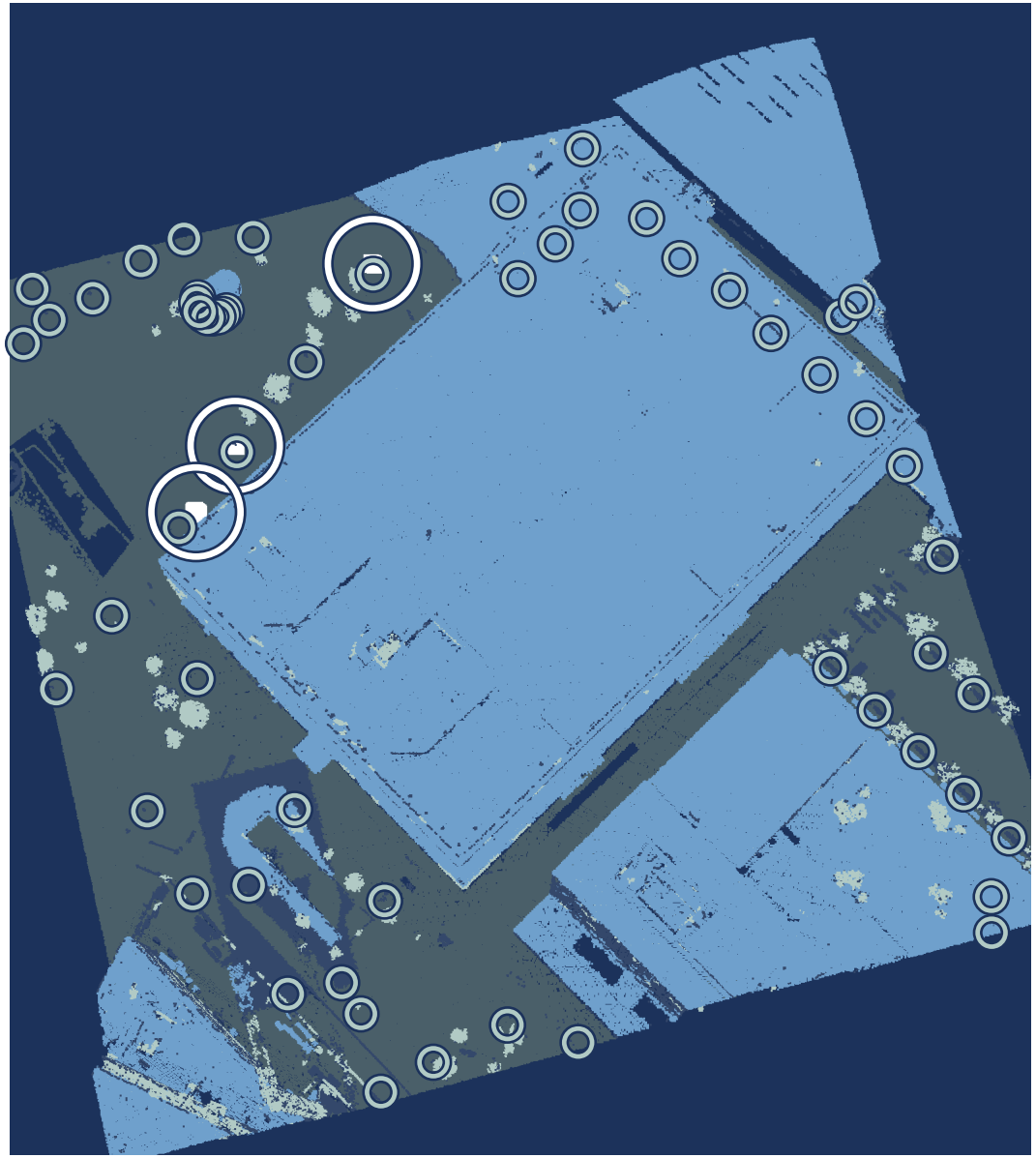
Pipeline labels



- Ground
- Buildings
- Vegetation
- Street furniture
- Unlabelled

722 961 points in 1.0 s, one run.
Ground vs AHN: 98.4 % precision, 94.4 % recall.

Data: AHN5, Het Waterschapshuis, CC BY 4.0.



Expectation vs reality: street furniture

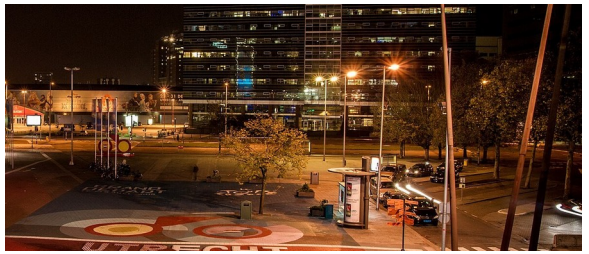
Expectation: the BGT register

61 poles 51 lamp posts, 10 bollards

Reality: our pipeline

3 objects 2 of them on a lamp post



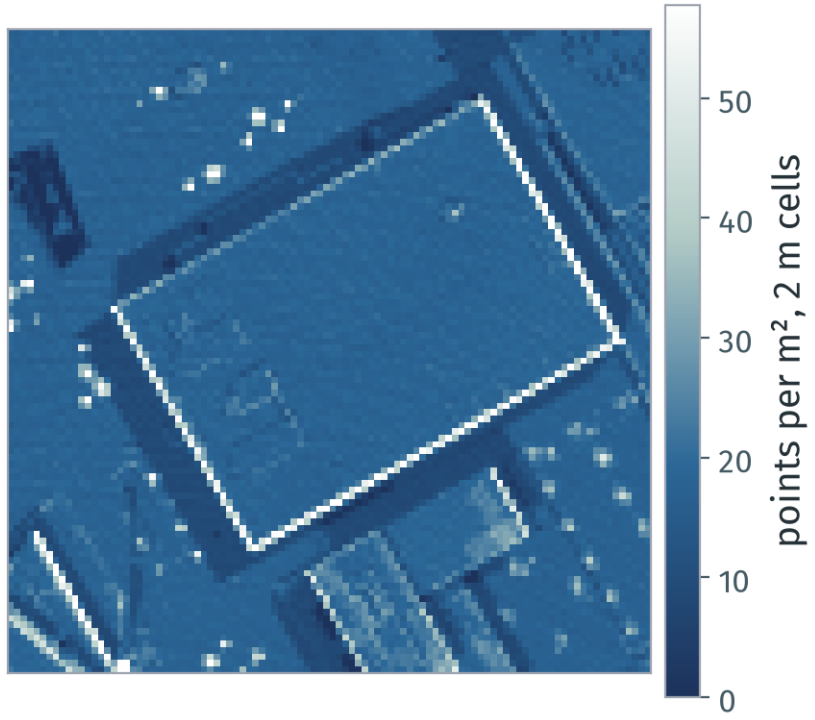


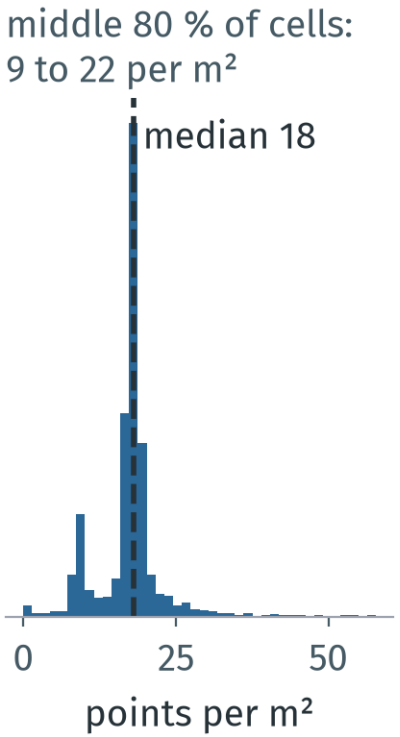
42 of 61 poles get no LiDAR point above 1.5 m.

Poles: BGT, Kadaster, CC BY 4.0. Photos: Wikimedia Commons, CC BY-SA 4.0 and CC BY 3.0.

Data: AHN5, Het Waterschapshuis, CC BY 4.0.

A point cloud samples the surface unevenly





Density, noise and gaps vary within one file. Measure them before tuning anything.

- Here the middle 80 % of cells hold 9 to 22 points per m²; median 18.
- Occlusion, flight-strip overlap, angle and material decide where points land.
- Every point carries error: AHN4 specifies at most 5 cm in height; AHN5 publishes no figure yet.

Quiz: which is faster, and by how much?

Distance of every point to the ground plane, 722 961 points.

A

```
a, b, c, d = plane
h = np.empty(len(P))  # distances
for i, p in enumerate(P):
    x, y, z = p[0], p[1], p[2]
    h[i] = abs(a*x + b*y + c*z + d)
```

B

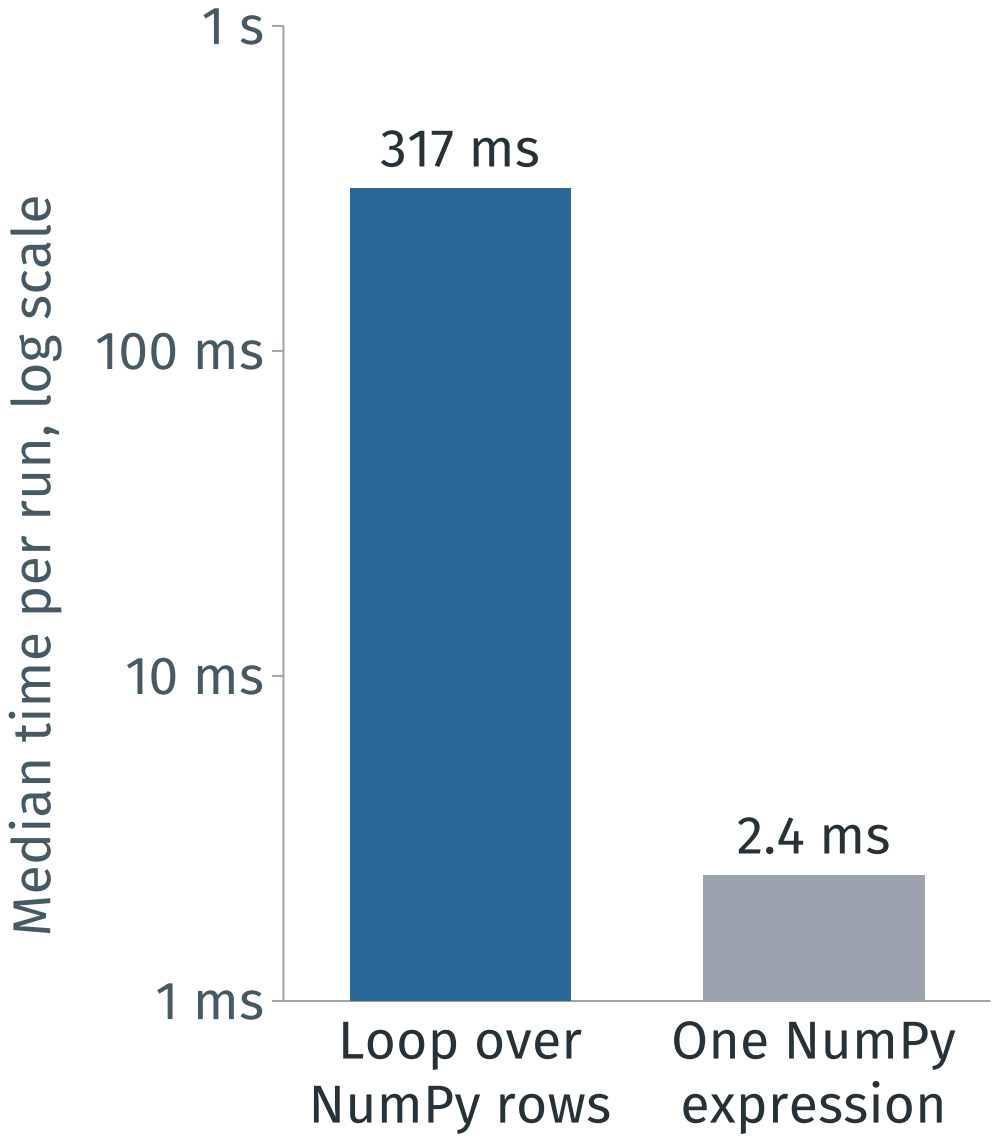
```
a, b, c, d = plane
x, y, z = P.T  # columns, no copy
h = np.abs(a*x + b*y + c*z + d)
```

The faster one wins by about: 2× | 10× | 100× | 1000×

Both give identical results (np.array_equal). P: the crop's xyz from slide 12, column-ordered. plane: one seeded segment_plane on the whole crop.

How much slower is the loop over rows?

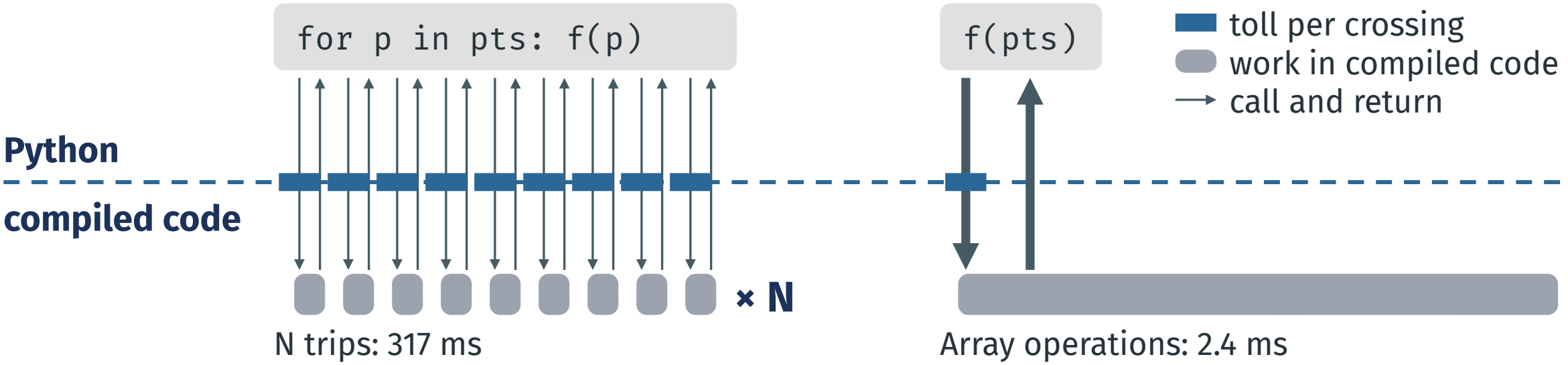
130× slower



Same data, same answer, far fewer calls into compiled code.

Conditions: Ryzen 7 5700X, Python 3.13.15, NumPy 2.5.3; 722 961 points, median of 7 runs; AHN5 crop (CC BY 4.0). Full: results/bench_cliff.json.

Batch calls into compiled code



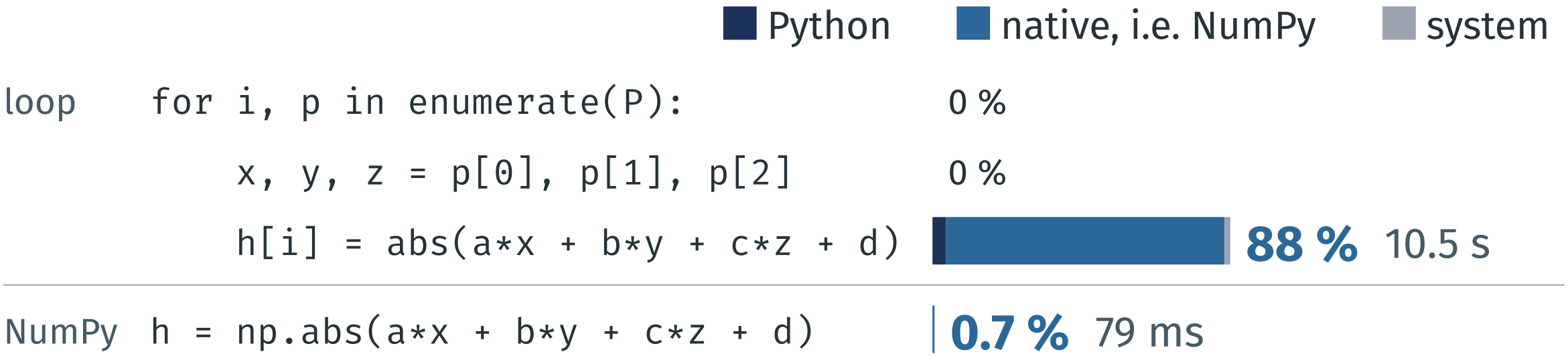
439 ns per full loop step

Prefer one call on the whole array over N calls on single points.

Conditions: Ryzen 7 5700X, Python 3.13.15, NumPy 2.5.3; 722 961 points, median of 7 runs; AHN5 crop (CC BY 4.0). Full: results/bench_cliff.json.

Profile before you vectorise

```
$ scalene run --cpu-only loop.py
```

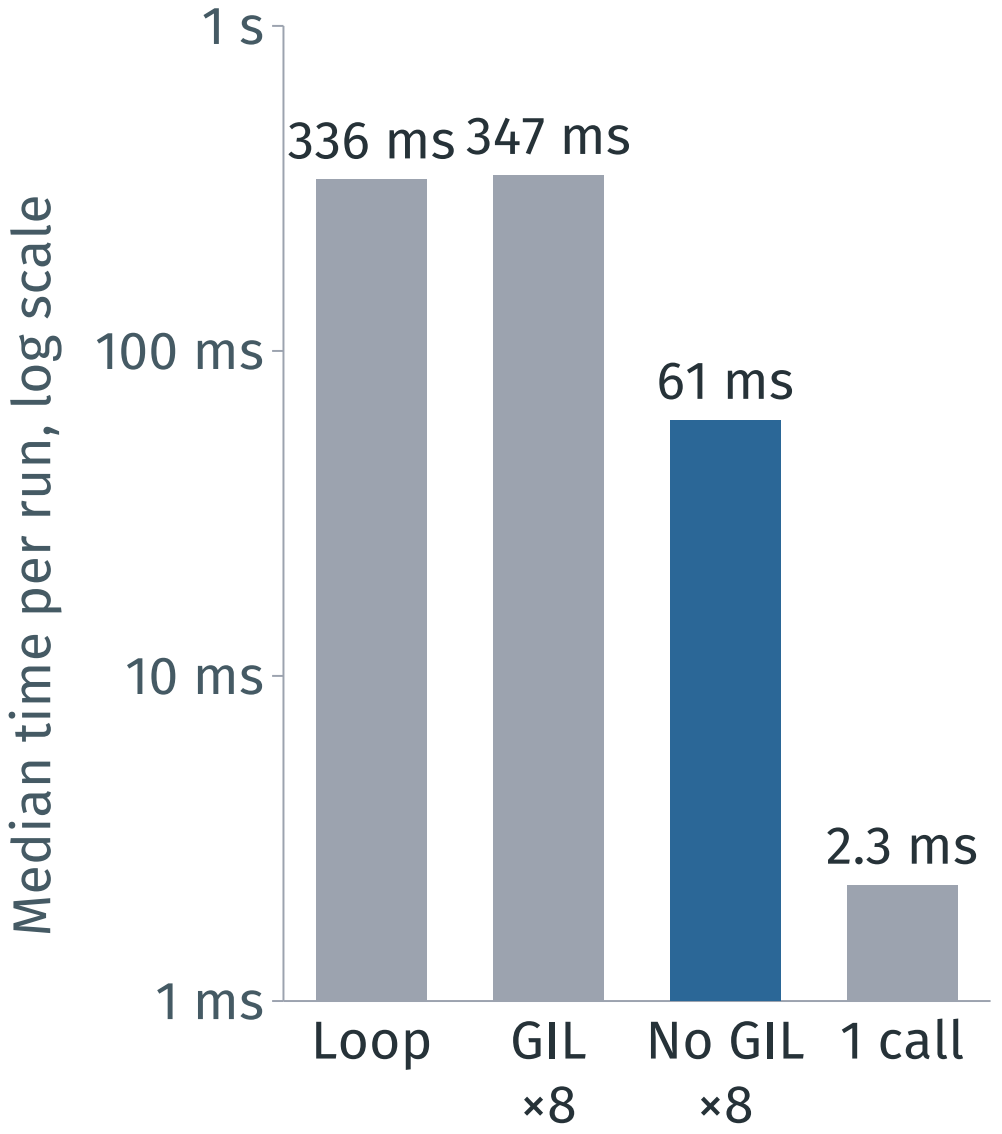


Times are totals over 30 calls of each version: about 349 ms and 2.6 ms per call under the profiler, against 317 ms and 2.4 ms timed alone (slide 22). Profile to find the line; benchmark to measure the gain.

Conditions: Ryzen 7 5700X, Python 3.13.15, Scalene 2.3.0 (--cpu-only, a sample every 1 ms); 722 961 points, both versions 30 times per run, median of 3 runs; AHN5 crop (CC BY 4.0). Shares are of the whole run, imports and reading the file included; times are share × the 11.9 s run. Full: results/profile_scalene.json.

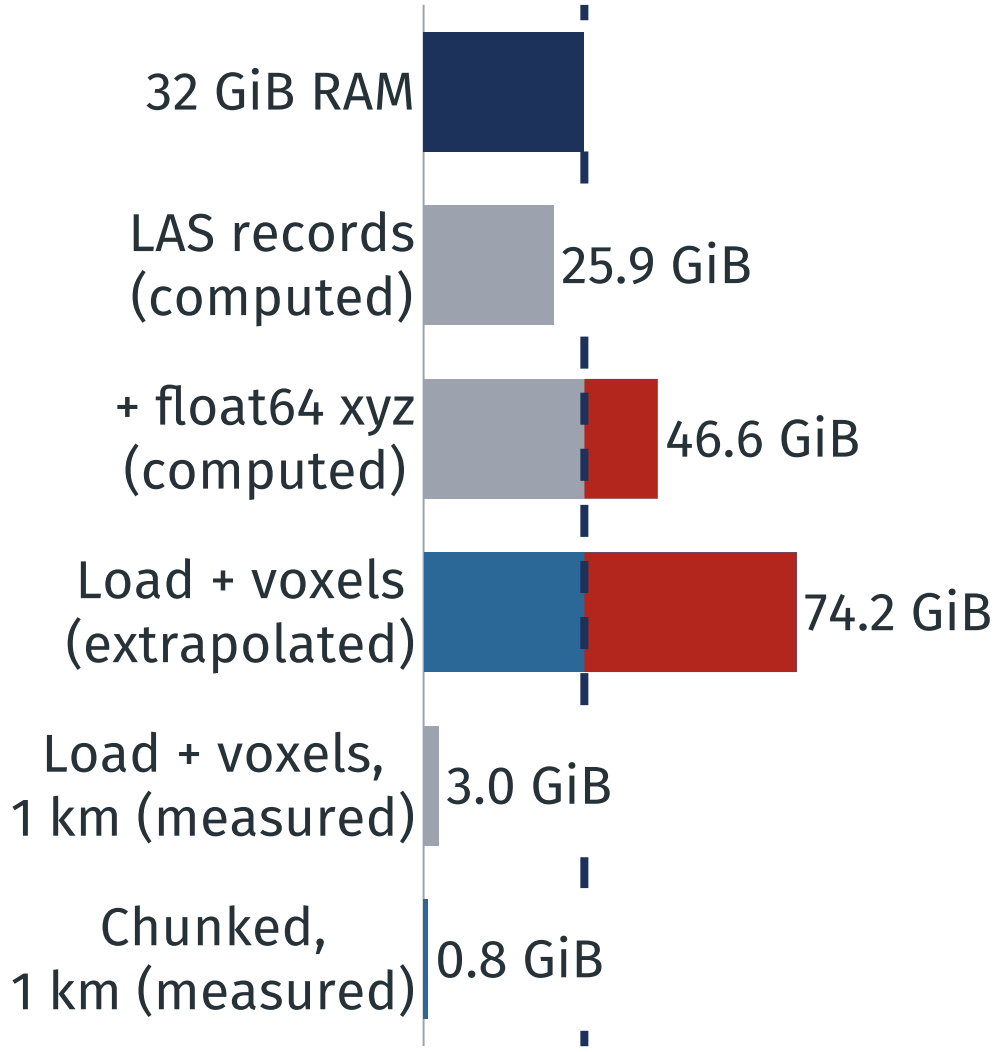
Will free-threaded Python save my loop?

5.5× faster on 8 free threads



×8: 8 threads. With the GIL, no gain. One NumPy call is 27× faster still.

Conditions: Ryzen 7 5700X, CPython 3.14.7 free-threaded and 3.14.7, NumPy 2.5.4; 722 961 points, median of 7 runs; AHN5 crop (CC BY 4.0). Full: results/bench_freethreaded.json.



Loading this area's AHN file outgrows 32 GiB

86 B retained per input point

926 203 338 points × ≈86 B each.

Big-file bars are computed, never run: load + voxels, arrays kept, temporaries excluded (with them: 133.1 GiB). The 1 km bars are measured peaks on one smaller file.

Stream chunks to fit within RAM.

Conditions: Ryzen 7 5700X, Python 3.13.15, NumPy 2.5.3; 21 013 926 points, median of 3 runs, each in a fresh process; AHN5 1 km file (CC BY 4.0). 1 GiB = 2^{30} B. Full: results/memory.json.

Take the four rules home

1 Think in arrays, not points

722 961 × 3, as views

slides 10, 12

2 Look before you compute

61 poles on the map, 3 found

slides 19, 20

3 Make it small first

3.0 GiB → 0.8 GiB peak

slides 13, 26

4 Let compiled code loop

130× faster than a loop

slides 21 to 25

Thank you

To the Python community and to everyone here today.



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github.com/Sloppy1999

Data: AHN5, Het Waterschapshuis, CC BY 4.0.
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